Domestication as Field-Specified Developmental Arrest

A Unified Causal Mechanism for the Domestication Syndrome Derived from Attractor Geometry,

Confirmed Against Five Independent Literature Sources, and Generating Novel Experimental Predictions

OrganismCore Framework — Attractor Geometry Series

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March 18, 2026

DERIVATION NOTICE — READ FIRST

The causal mechanism proposed in this paper was derived from first principles using the attractor geometry framework before any literature search was conducted. The predictions were locked in a timestamped repository record (OrganismCore, 2026-03-18) prior to the literature verification reported in Section 5. The confirmations against Marcström & Essen-Möller (1968), Hemmer (1990), Wilkins et al. (2014), Belyaev (1979), and the feralization sequence literature are therefore confirmations of pre-specified predictions, not the source of those predictions. This eliminates post-hoc pattern matching as an explanation for the correspondence.

Repository record: github.com/Eric-Robert-Lawson/attractor-oncology

Pre-registration of novel experimental predictions (P7–P11): deposited Zenodo, 2026-03-18.

Abstract

The domestication syndrome — the convergent suite of morphological and behavioural traits appearing in every domesticated species including floppy ears, reduced pigmentation, shorter snout, smaller adrenal glands, and retained juvenile behaviour — has resisted unified causal explanation since its systematic description. Wilkins, Wrangham and Fitch (2014) correctly identified the cellular mechanism: mild reduction in neural crest cell migration. They did not identify the causal origin of that reduction. Here we propose and defend that origin: the domestication coherence gradient field (N_{domestic}) suppresses the maternal hypothalamic-pituitary-adrenal (HPA) axis, alters the intrauterine coherence gradient environment, and arrests neural crest cell migration at the juvenile-proximate developmental stage. Domestication is therefore not genetic modification of the organism. It is field-specified developmental arrest of the neural crest migration program. The domestic phenotype is the phenotypic signature of that arrest. Feralization is developmental resumption when the maintenance field is removed. This mechanism was derived from the attractor geometry framework ($S + N + G \to R$) before literature consultation and is confirmed against five independent literature sources. The mechanism predicts a specific asymmetry in domestication vs. feralization rates, a specific reversion sequence in feral animals, a specific cross-species universality of the syndrome, and a specific set of novel experimental outcomes (P7–P11) not predicted by the genetic model. The mechanism is structurally identical to EZH2-mediated attractor arrest in triple-negative breast cancer and nitrogen starvation-mediated arrest in the LECA protocol, suggesting a scale-invariant field-arrest mechanism operative from single cells to whole organisms.

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1. Introduction: The Open Causal Question

Domestic animals are morphologically and behaviourally distinct from their wild ancestors in ways that are strikingly convergent across species. The dog differs from the wolf, the domestic pig from the wild boar, the domestic cat from the African wildcat, and the domesticated silver fox from its wild counterpart in a predictable, overlapping set of traits that has been termed the *domestication syndrome* [1]: floppy ears, patches of depigmentation, shortened snout, smaller adrenal glands, reduced stress reactivity, docility, and retention of juvenile behaviour into adulthood (neoteny).

The cross-species universality of this syndrome is remarkable. These species share no recent common ancestor. Their domestication events were geographically separated and historically independent. Yet the phenotypic outcome is the same. This convergence demands a unified causal explanation.

Wilkins et al. (2014) provided the most significant advance: a unified cellular mechanism. They demonstrated that all traits of the domestication syndrome map onto a single developmental perturbation — mild reduction in neural crest cell (NCC) migration during embryogenesis. Neural crest cells are multipotent embryonic cells that migrate from the dorsal neural tube to populate ear cartilage, craniofacial structures, melanocytes, adrenal medulla, and components of the peripheral nervous system. When NCC migration is mildly reduced, all syndrome traits appear simultaneously. The cellular mechanism is correct and well-supported.

What Wilkins et al. did not provide is the causal origin of the NCC migration reduction. Their explanation invokes genetic change — alleles affecting NCC number, migration capacity, or differentiation, co-selected during domestication alongside tameness alleles. This genetic explanation is incomplete for three reasons:

1. **Frequency:** Marcström & Essen-Möller (1968) documented domestication syndrome traits — piebald coat, drooping ears, skull shape changes — in the first captive generation of wild boar *without selective breeding*, at frequencies too high for recessive allele expression to explain.
2. **Asymmetry:** Feralization (domestic → wild) occurs in months to one generation. Redomestication (wild → domestic) requires many generations of sustained selection. This asymmetry is not predicted by genetic models in which allele frequencies shift at equivalent rates under equivalent selection.
3. **Sufficiency:** The Belyaev fox experiment [2] demonstrated that selecting only for HPA axis suppressibility (tameness) produces the full domestication syndrome without morphology selection. The genetic model requires co-selection of linked alleles; the field model requires only that HPA suppression alters the intrauterine coherence gradient.

We propose that the causal origin of NCC migration reduction is not genetic but *field-specified*: the domestication coherence gradient field acts on the maternal HPA axis, which alters the intrauterine developmental environment, which changes the coherence gradients guiding NCC migration, which arrests that migration at the juvenile-proximate stage. The domestic phenotype is the organismal signature of that field-specified developmental arrest.

2. The Attractor Geometry Framework

2.1. The Complete Developmental Equation

The standard model of developmental biology is:

$$\text{Genome} \rightarrow \text{Organism}$$

This model is incomplete. The complete model, as formalised in the attractor geometry framework [7, 8, 9], is:

$$S + N + G \rightarrow R$$

where S is the structural substrate (the genome and its encoded build capacity), N is the external coherence gradient field (the environmental specification of which developmental attractor the genome resolves to), G is the developmental geometry (the global attractor landscape, basin structure, and current position of the system in state space), and R is the resulting phenotype.

This formulation follows directly from the Waddington epigenetic landscape [10], whose mathematical formalisation as a quasi-potential function:

$$U(\mathbf{x}) = -k_B T \ln P_{\text{ss}}(\mathbf{x})$$

(where P_{ss} is the steady-state probability distribution over developmental state space) was provided by Wang et al. (2014) [9]. The landscape U is not readable from the gene regulatory network F alone without computing the global state space geometry. This is the mathematical basis for the incompleteness of the genome-only model: the genome encodes F; the phenotype is determined by U; and U is shaped by both F and the external field N.

2.2. Attractor Basins and Developmental Arrest

In the Waddington landscape, stable phenotypes correspond to attractor basins — valleys in U whose depth reflects the stability of the corresponding developmental state. A system placed in a coherence gradient field N will navigate to the basin whose geometry is specified by N.

Developmental arrest occurs when a coherence gradient field holds a developing system in a shallower, earlier-stage basin rather than allowing the developmental trajectory to proceed to its deep-basin endpoint. The arrested state is phenotypically stable as long as the maintenance field is applied. When the maintenance field is removed, the system exits the shallow basin and continues its developmental trajectory toward the nearest deep basin.

This mechanism has been directly demonstrated at multiple scales:

- *Single cell:* Nitrogen starvation arrests yeast in the LECA-proximate attractor basin. Restore nitrogen; the modern eukaryotic program resumes [17].
- *Single cell:* EZH2-mediated H3K27me3 deposition silences FOXA1 and arrests breast epithelial cells in the cancer attractor basin. Tazemetostat (EZH2 inhibitor) de-arrests the cell; the luminal program resumes [16].
- *Organism:* Inversion of the coherence gradient field (mineral water above, UV-A below) arrests the radish developmental program in the inverted architectural attractor. The root grows upward; the shoot grows downward. Same genome; inverted field; inverted organism [18].

We propose that domestication is the same mechanism at the organismal scale: a field-specified developmental arrest of the neural crest migration program.

3. The Domestication Mechanism: Field-Specified Neural Crest Arrest

3.1. The Domestication Coherence Gradient Field

The domestication coherence gradient field N_{domestic} is defined by the following components acting simultaneously:

- **Predation removal:** Elimination of fight-or-flight activation. Sustained HPA axis suppression.
- **Ad libitum food provision:** Elimination of metabolic stress. Sustained anabolic state.
- **Temperature regulation:** Elimination of thermoregulatory stress load.
- **Social stability:** Elimination of dominance-hierarchy conflict. Oxytocin-dominant social environment.
- **Human contact from early development:** Imprinting on the human social field from the critical window.

This field does not modify the genome. It modifies the physiological state of the animal occupying it — specifically the HPA axis set point and the resulting hormonal environment of any gestating offspring.

3.2. The Causal Chain

The complete causal chain from field to phenotype is:

N_{domestic}
↓
Maternal HPA axis suppression (reduced CRH, ACTH, cortisol)
↓
Reduced intrauterine glucocorticoid exposure
↓
Altered embryonic coherence gradients
(BMP4/7, Wnt1/3a, CXCL12/CXCR4, EDNRB, FGF8)
↓
Partial arrest of neural crest cell migration
(juvenile-proximate stage)
↓
Basin B occupation (domestic attractor)
↓
Domestication syndrome
(floppy ears, depigmentation, smaller adrenal,
shorter snout, docility, juvenile behaviour retention)

Each step in this chain has independent support in the developmental biology and stress physiology literature:

Step 1 (HPA suppression): Glucocorticoids are known regulators of NCC migration. Elevated cortisol suppresses NCC proliferation and increases apoptosis in migrating NCC populations [12]. N_{domestic} chronically suppresses maternal cortisol.

Step 2 (Intrauterine gradient alteration): Maternal glucocorticoid levels directly influence foetal HPA axis programming via placental 11β-HSD2 activity and direct cortisol transfer. The intrauterine hormonal environment shapes the coherence gradients for embryonic NCC migration [13].

Step 3 (NCC gradient dependence): NCC migration is not genetically autonomous. It is guided by chemokine gradients (CXCL12/CXCR4), BMP gradients, Wnt signalling, and extracellular matrix composition — all of which are sensitive to the hormonal environment of the embryo [11].

Step 4 (Syndrome as arrest signature): Wilkins et al. (2014) confirmed that all syndrome traits are downstream of NCC migration reduction. Our contribution is the causal origin of that reduction: it is field-specified, not allele-specified.

3.3. Why NCC Migration Is a Developmental Trajectory, Not a Genetic Switch

Neural crest cell migration has a starting state (NCC cells at the dorsal neural tube), a directed trajectory (migration through embryonic tissue guided by coherence gradients), and an endpoint (full differentiation into all NCC-derived tissues).

Full migration to completion = Basin A = wild-type phenotype.

Arrested migration at early stage = Basin B = domestic phenotype.

The migration is not a binary genetic switch between two allelic states. It is a continuous developmental trajectory that can be held at any intermediate stage by the coherence gradient environment of the developing embryo. The domestication field holds it at the juvenile-proximate stage. The wild field allows it to proceed to completion.

This is the precise mechanism of developmental arrest. The domestic pig is not a genetically different organism from the wild boar. It is the same organism whose developmental trajectory was arrested at an earlier stage and held there by field maintenance.

4. The Two-Basin Landscape of *Sus scrofa*

The attractor landscape of *Sus scrofa* contains (at minimum) two primary developmental basins:

	Basin A (Wild Boar)	**Basin B (Domestic Pig)**
Depth	Deep. Carved by millions of years of	
natural selection.	Shallow. Carved by ∼10,000 years	
of domestication selection.		
NCC migration	Runs to completion.	Arrested at juvenile-proximate stage.
Field requirement	N_{wild}: predation, food	
scarcity, thermoregulatory challenge.	N_{domestic}: protection,	
ad libitum food, stability.		
Phenotype	Coarse hair, erect ears, elongated	
snout, large adrenal, aggressive,		
lean, tusked, camouflaged.	Fine hair, floppy ears, shortened	
snout, small adrenal, docile, fat-depositing, depigmented.		
Maintenance	Self-maintaining under	
N_{wild}.	Requires continuous	
N_{domestic} application.		

The critical implication of the basin depth asymmetry is the **transition rate asymmetry**:

- **Basin B → Basin A (feralization):** Remove N_{domestic}. The system exits the shallow Basin B and rolls into the deep Basin A. Rate: months to one generation. Energy barrier: low.
- **Basin A → Basin B (domestication):** Must carve Basin B against the restoring force of Basin A. Requires sustained, directed N_{domestic} application over many generations. Energy barrier: high.

This asymmetry is geometrically necessary from the landscape structure. It is not explained by the genetic model, in which allele frequencies shift at approximately equivalent rates under equivalent selection pressure in both directions.

5. Confirmation Against Existing Literature

The following confirmations are against literature identified *after* the mechanism was derived. Each confirmation is of a prediction entailed by the field-arrest model.

5.1. Marcström & Essen-Möller (1968): The Buried Confirmation

Marcström and Essen-Möller placed wild-caught *Sus scrofa* in captivity without selective breeding. In the first captive generation, the following appeared spontaneously:

- White piebald patches (melanocyte migration arrest — NCC-derived)
- Widening and drooping of ears (cartilage reduction — NCC-derived)
- Skull shape changes toward shorter, wider craniofacial profile (NCC-derived)
- Curly tails

The critical point: these traits appeared without selective breeding, in animals with fully wild genomes, simply by applying N_{domestic}. The frequency of trait expression was too high for recessive allele expression to explain. **Confirmed by field-arrest model. Not explained by genetic model.**

5.2. Hemmer (1990): Adrenal Reduction as Earliest Change

Hemmer's systematic analysis documented that adrenal gland size reduction is among the earliest and most consistent morphological changes in captive wild animals across species [4]. The adrenal medulla is NCC-derived. Its size reflects the degree of NCC migration completion to the adrenal anlage.

Prediction confirmed: the fastest-responding tissue is the one whose developmental program is most directly downstream of the field's primary action (HPA axis suppression). **Confirmed.**

5.3. Wilkins, Wrangham & Fitch (2014): The Cellular Mechanism Identified

Wilkins et al. correctly identified that all domestication syndrome traits map to a single developmental perturbation: mild reduction in NCC migration. They attributed this to genetic change. We provide the causal origin of the perturbation: field-specified alteration of the intrauterine NCC migration guidance environment via maternal HPA axis suppression.

Our paper completes the causal chain they left open. It does not contradict their finding. It explains why it is true.

5.4. Belyaev (1979) and Trut et al. (2009): Tameness Is Not the Cause

The silver fox domestication experiment selected only for HPA axis suppressibility (reduced fear/flight response). Within 10–15 generations, the full domestication syndrome appeared: floppy ears, piebald coat, shortened snout, curly tail, reduced adrenal size, juvenile behaviour retention.

Standard interpretation: co-selected genetic alleles linking tameness to morphology.

Field-arrest interpretation: selecting for the most HPA-suppressible individuals selected for those whose intrauterine environment was closest to N_{domestic}. Their offspring's NCC migration was most arrested. The syndrome appeared not because tameness alleles are linked to ear alleles. It appeared because both are downstream of the same field-specified arrest mechanism.

Tameness is the behavioural signature of Basin B occupation. The morphological traits are the structural signature of the same Basin B occupation. They appear together because they are both downstream of the same arrested developmental trajectory.

5.5. The Feralization Sequence: A Reversion Signature

In feralization, morphological reversion occurs in a specific sequence: behaviour first (days to weeks), then hair coarseness (weeks to months), then body composition (months), then craniofacial structure (requires offspring generation for full expression).

Prediction from field-arrest model: The reversion sequence should follow the order of developmental plasticity from highest to lowest — epigenetically-regulated traits (fastest) to structurally-embedded traits (slowest). This is exactly the observed sequence. The genetic model predicts no specific sequence, as allele frequencies shift at equivalent rates across traits. **Confirmed.**

5.6. Summary of Confirmations

Prediction (derived before literature)	Source	Status
Domestication syndrome in F1 captive wild animals without selective breeding	Marcström (1968)	**Confirmed**
Trait frequency too high for recessive allele explanation	Marcström (1968)	**Confirmed**
Adrenal size reduction earliest and most consistent change	Hemmer (1990)	**Confirmed**
All syndrome traits → single developmental perturbation (NCC)	Wilkins et al. (2014)	**Confirmed**
Tameness selection alone produces full syndrome	Belyaev (1979)	**Confirmed**
Feralization rate asymmetry vs. re-domestication	Multiple sources	**Confirmed**
Reversion sequence tracks developmental plasticity order	Feralization literature	**Confirmed**

6. Novel Experimental Predictions

The following predictions are entailed by the field-arrest model and are not entailed by the genetic model. They constitute the experimental program that distinguishes the two frameworks.

NOTE ON PRE-REGISTRATION: Predictions P7–P11 are pre-registered on Zenodo (2026-03-18) prior to any experimental data collection. DOI to be inserted upon Zenodo confirmation.

6.1. P7 — One-Generation Morphological Re-domestication via Cross-Fostering

Design: F1 offspring of wild-caught *Sus scrofa* (fully wild genome, confirmed by genotyping) placed at birth with domestic foster sow under full N_{domestic} from birth. Genotype-matched controls raised under N_{wild}.

Measurements at 3, 6, 12 months: adrenal gland size (at necropsy), cortisol response to standardised stressor, ear cartilage stiffness and position, coat pigmentation pattern (% piebald surface area), hair coarseness, snout length/width ratio.

Field-arrest prediction: N_{domestic}-raised F1 will show smaller adrenal glands and reduced HPA reactivity compared to genotype-matched N_{wild}-raised F1. Ear and pigmentation changes will be intermediate (postnatal field only; full expression requires intrauterine field application — see P8).

Genetic model prediction: No morphological difference between conditions with identical genomes.

The models diverge. The experiment has not been done.

6.2. P8 — Maternal Field Window Dependence

Design: Feral sow inseminated and held under N_{domestic} from conception. F1 offspring raised identically post-birth. Compare to P7 controls (field applied only from birth).

Prediction: F1 offspring of sow in N_{domestic} from conception will show larger morphological shift toward domestic phenotype than F1 offspring with only postnatal N_{domestic} exposure, with identical genomes.

The gradient of morphological shift across conditions (full prenatal + postnatal > postnatal only > wild control) specifies the NCC migration critical window relative to parturition.

6.3. P9 — H3K27me3 Enrichment at NCC Migration Loci in Domestic vs. Wild Embryos

Design: H3K27me3 ChIP-seq on embryonic tissue (E14–E18, NCC active migration stage) from domestic *Sus scrofa* and wild boar.

Prediction: Domestic pig embryos will show elevated H3K27me3 (EZH2-mediated silencing) at NCC migration driver loci relative to wild boar embryos at the same developmental stage, specifically:

- BMP4, BMP7 (NCC migration guidance)
- WNT1, WNT3A (NCC specification)
- CXCL12, CXCR4 (directional migration)
- EDNRB (pigment cell migration)
- SOX10, SNAI2 (NCC identity maintenance)

Significance: This would establish molecular identity between the domestication arrest mechanism and EZH2-mediated cancer attractor arrest — connecting domestication biology directly to the cancer biology framework.

6.4. P10 — Breed Domestication Depth Predicts Feralization Rate

Prediction: Feralization rate (months to morphological reversion threshold) correlates negatively with domestication depth (genetic distance from wild boar; generations of intense selection).

Ranking (slowest to fastest feralization): Large White/Landrace (most derived) > heritage breeds > Mangalitsa/Ossabaw (least derived, closest to wild boar ancestry).

Design: Comparative morphological measurement (hair coarseness, adrenal weight, HPA reactivity) in feral populations derived from known breed starting points at 3 and 6 months post-feralization.

6.5. P11 — EZH2 Inhibition Shifts F1 Morphology Toward Wild-Type

Rationale: If H3K27me3 at NCC migration loci maintains the domestication arrest (P9), then EZH2 inhibition in a pregnant domestic sow should partially de-repress NCC migration drivers and shift F1 morphology toward wild-type.

Design: Pregnant domestic sows treated with sub-therapeutic tazemetostat (EZH2 inhibitor, FDA-approved) during the NCC migration window. F1 morphology assessed at 6 months vs. vehicle-treated controls.

Prediction: F1 offspring of EZH2-inhibited dams will show increased hair coarseness, larger adrenal glands, and higher HPA reactivity relative to F1 offspring of vehicle-treated dams. **No genetic modification. Same genome. Field perturbation shifts phenotype.**

7. The Cross-Scale Mechanistic Identity

The domestication arrest mechanism is not unique to Sus scrofa or to organismal-scale biology. The same mechanism — field-specified arrest of a developmental trajectory at an attractor basin maintained by epigenetic silencing, reversible by field change — is operative at multiple scales:

Scale	System	Arrest field	Resumption	Reference
Single cell	TNBC	EZH2/H3K27me3 silences FOXA1	Tazemetostat removes silencing	[16]
Single cell	LECA yeast	Nitrogen starvation arrests modern program	Restore nitrogen	[17]
Organism	Radish	Standard field specifies root-down attractor	Invert field → root-up	[18]
Organism	Sus scrofa	N_{domestic} arrests NCC migration	Remove field → feralization	This paper

The structural identity across scales is not metaphorical. The same causal operation is occurring at each scale: the external coherence gradient field N specifies which attractor basin the structural substrate S resolves to within the developmental geometry G, producing result R. The genome or cellular build capacity never changes. The field changes. The phenotype changes.

If the domestication arrest mechanism involves EZH2-mediated H3K27me3 at NCC loci (P9), then the molecular mechanism is identical to the cancer arrest mechanism. Tazemetostat, the FDA-approved EZH2 inhibitor currently in clinical use for cancer, would be the reagent that de-arrests the domestication program (P11). The same drug. The same molecular mechanism. Cancer and domestication as instances of the same field-specified epigenetic arrest. Different scale. One mechanism.

8. Why the Genetic Model Is Incomplete

The genetic model of domestication correctly explains:

- Why domesticated traits are heritable (the alleles and epigenetic states are transmitted)
- Why selective breeding accelerates domestication (selection deepens Basin B)

The genetic model cannot explain:

1. **Cross-species syndrome universality:** Every domesticated species, regardless of phylogenetic distance, shows the same syndrome. Convergent acquisition of linked NCC-affecting alleles in every independent domestication event is implausible. A shared field type producing a shared arrest is parsimonious.
2. **First-generation appearance without selection:** Marcström (1968) documented syndrome traits in F1 captive wild boar without selection. Recessive allele expression cannot produce high-frequency, convergent trait appearance in F1 without selection.
3. **Feralization/domestication rate asymmetry:** Allele frequencies shift at equivalent rates under equivalent selection. Feralization in months vs. domestication in generations is not explained. Basin depth asymmetry explains it directly.
4. **Syndrome from tameness selection alone:** The Belyaev result requires either tight genetic linkage between tameness and every morphological trait simultaneously, or a single upstream mechanism (HPA suppression → intrauterine coherence gradient → NCC arrest) producing all traits simultaneously. The field-arrest model provides this upstream mechanism.
5. **Reversion sequence:** Allele frequencies change at equivalent rates across traits. The ordered reversion sequence in feralization (behaviour → hair → body composition → craniofacial) has no explanation under the genetic model. It is geometrically necessary under the developmental arrest model.

9. Discussion

9.1. What Domestication Actually Is

Domestication, under the field-arrest model, is not the genetic modification of wild animals into domestic ones. It is the engineering of a coherence gradient field that arrests the developmental trajectory of wild animals at the juvenile-proximate basin and holds them there by continuous field maintenance.

The genome does not change. The field changes. The developmental trajectory is arrested. The organism expresses the phenotypic signature of that arrest.

This has two immediate practical implications:

Implication 1: The wild phenotype is always recoverable. It is not lost. It is suspended. Remove the maintenance field and the developmental trajectory resumes. This is why feralization is fast and universal.

Implication 2: The domestication field, not the genome, is the primary engineering target. To produce a domestic phenotype from a wild starting point, the correct intervention is not selective breeding per se — it is engineering the coherence gradient field that will arrest NCC migration at the domestic-proximate stage in the developing offspring.

9.2. The Evidence That Has Always Been There

Marcström and Essen-Möller published their captive wild boar data in 1968. The results were correctly observed: domestication syndrome traits in first-generation captive animals without selective breeding. They were incorrectly interpreted: latent allele expression under relaxed selection.

The correct interpretation has been unavailable because the framework to state it — field-specified developmental arrest as a causal mechanism in vertebrate biology — did not exist. The evidence was always there. The framework to read it was not.

9.3. The Wojtek Problem

Wojtek, the Syrian brown bear who served with the Polish Army (1943–1963), was raised from cubhood in the human social field. He became fully socialised, learned to carry ammunition, ate cigarettes, and lived as a functional member of a human military unit. Standard interpretation: bears are intelligent and can be trained.

Geometric interpretation: Wojtek was placed in N_{domestic} during the critical developmental window for HPA axis calibration and social field imprinting. His developmental program resolved to the domestic-proximate basin not because he was trained but because that was the basin specified by the field in which he developed.

Every wildlife rehabilitation programme that rescues neonates is running this experiment continuously and observing this result consistently. The result is interpreted as taming or socialisation. It is developmental arrest. The difference is not semantic. It is causal.

9.4. Relation to Cancer Biology and the LECA Protocol

The field-arrest model of domestication connects directly to two active experimental programs:

The FOXA1/EZH2 work in triple-negative breast cancer proposes that cancer is a cellular developmental arrest — the cell held in an ancestral attractor basin by EZH2-mediated epigenetic silencing of the identity program. If P9 confirms H3K27me3 enrichment at NCC loci in domestic pig embryos, the same molecular mechanism maintains developmental arrest at two scales simultaneously: cell and organism.

The LECA protocol proposes that the precambrian ionic coherence gradient field specifies the LECA attractor basin in living yeast. The domestication protocol is the same operation at the organismal scale: the domestication coherence gradient field specifies the domestic attractor basin in *Sus scrofa*. Both are recoverable. Both are field-specified. Both involve the same genome operating in a different field.

The pig, the cancer cell, the yeast, and the radish are not separate biological curiosities. They are the same causal geometry at different scales. The domestication paper and the cancer paper are the same paper at different scales.

10. Conclusions

We propose that domestication is field-specified developmental arrest of the neural crest migration program. The domestication coherence gradient field suppresses the maternal HPA axis, alters the intrauterine coherence gradient environment, and arrests neural crest cell migration at the juvenile-proximate developmental stage. The domestication syndrome is the phenotypic signature of that arrest. Feralization is developmental resumption when the maintenance field is removed.

This mechanism:

1. Provides the causal origin of the NCC migration reduction identified by Wilkins et al. (2014)
2. Explains the cross-species universality of the domestication syndrome without invoking convergent genetic change
3. Explains the first-generation appearance of syndrome traits in captive wild animals without selection (Marcström 1968)
4. Explains the asymmetry between feralization rate and re-domestication rate from basin depth geometry
5. Explains the ordered reversion sequence in feralization from developmental plasticity principles
6. Generates five novel experimental predictions (P7–P11) that are not entailed by the genetic model and that constitute a decisive test between frameworks
7. Is structurally identical to EZH2-mediated attractor arrest in cancer and nitrogen starvation arrest in the LECA protocol, suggesting a scale-invariant field-arrest mechanism across vertebrate biology

The domestic pig is not a genetically modified wild boar. It is a wild boar whose developmental trajectory was arrested at the juvenile-proximate basin by a field that has been continuously maintained for $\sim$10,000 years. The wild boar was never lost. It was suspended. Remove the field and it resumes.

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Document ID: DOMESTICATION_AS_DEVELOPMENTAL_ ARREST_v1.0
Date: 2026-03-18

Author: Eric Robert Lawson / OrganismCore

Derivation record: `github.com/Eric-Robert-Lawson/attractor-oncology`

Pre-registration of P7–P11: Zenodo, 2026-03-18 (DOI to be inserted upon confirmation)

Target journal: *BioEssays* (primary); *Genetics* (secondary — direct extension of Wilkins et al. 2014)

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